Definition :

Let A and B be two non empty sets then a subset 'f' of $A \times B$ is said to be a function from A to B if $\forall x \in A \ \exists$ unique $y \in B$.

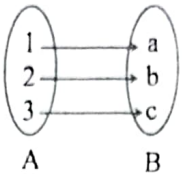
i.e. $f = \{ (x, y) \mid x \in A, y \in B \quad \forall x \in A \ \exists \text{ unique } y \in B \}$

Example $A = \{ 1, 2, 3 \}$

$B = \{ a, b, c \}$

$A \times B = \{ (1, a), (1, b), (1, c), (2, a), (2, b), (2, c), (3, a), (3, b), (3, c) \}$

1. Consider $f = \{ (1, a), (2, b), (3, c) \}$

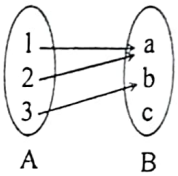


$\because f \subseteq A \times B$

& $\forall x \in A \ \exists$ unique $y \in B$

$\therefore$ f is a function.

2. Consider $f = \{ (1, a), (2, a), (3, b) \}$

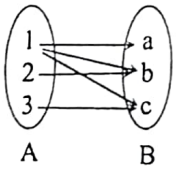


$\because f \subseteq A \times B$

& $\forall x \in A \ \exists$ unique $y \in B$

$\therefore$ f is a function.

3. Consider $f = \{ (1, a), (1, b), (1, c), (2, b), (3, c) \}$



$\because f \subseteq A \times B$

for $1 \in A$ there are three y i.e. $a, b, c \in B$

$\therefore$ It does not follow uniqueness.

$\therefore$ f is not a function.

Domain, Co-domain, Image & Range of a function :

Let $f : A \to B$ be a function, then

i) A is called domain and

ii) B is called codomain.

iii) $\because$ f is a function

$\therefore \forall x \in A \ \exists$ unique $y \in B$ then y is called f-image of x.

iv) A subset of co-domain is said to be range of f if it contains all images $\forall x \in A$

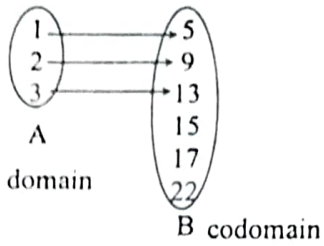
Consider, $A = \{1, 2, 3\}$

$B = \{5, 9, 13, 15, 17, 22\}$

and $f = \{(1,5), (2, 9), (3, 13)\}$

i) Injective

Consider $f(x_1) = f(x_2)$



$\because f \subseteq A \times B$

$f(1) = 5 = 4 \times 1 + 1$

$f(2) = 9 = 4 \times 2 + 1$

$f(3) = 13 = 4 \times 3 + 1$

$\therefore$ we can define a function $\forall x \in A$ as

$f(x) = 4x + 1$

Here $f(5)$ can not be determined $\because 5$ is not an element of A

$\because$ Range of f is the set of all images $\forall x \in A$

$\therefore$ Range (f) $= \{5, 9, 13\}$

Injective Function :

Let $f : A \to B$ be a function then f is said to be injective (one – one)

if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

or

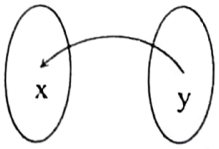
$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$

Surjective Function :

Let $f : A \to B$ be a function then f is said to be surjective (onto)

if $\forall y \in B$ (codomain) $\exists$ pre image $x \in A$ (domain)

or Range of f = codomain.



A (Domain) B (Codomain)

Bijective Function :

Let $f : A \to B$ be a function then f is said to be bijective if f is injective and surjective both.

Inverse Function :

Let $f : A \to B$ be a function then f is said to be invertible if $f^{-1} : B \to A$ is a function.

NOTE : A function $f : A \to B$ is invertible iff f is bijective.

Examples :

1. A function $f : R - \left\{\frac{7}{3}\right\} \to R - \left\{\frac{4}{3}\right\}$ is defined as

$$f(x) = \frac{4x - 5}{3x - 7}$$

Prove that 'f' is bijective and find the rule for f^{-1}.

Soln. :

$\because f : R - \left\{\frac{7}{3}\right\} \to R - \left\{\frac{4}{3}\right\}$

is defined as $f(x) = \dfrac{4x - 5}{3x - 7}$

i) Injective

Consider $f(x_1) = f(x_2)$

$$\therefore \quad \frac{4x_1-5}{3x_1-7} = \frac{4x_2-5}{3x_2-7}$$

or $\quad (4x_1-5)(3x_2-7) = (3x_1-7)(4x_2-5)$

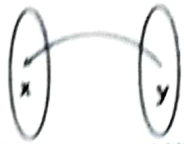
or $\quad 12x_1x_2 - 28x_1 - 15x_2 + 35 = 12x_1x_2 - 28x_2 - 15x_1 + 35$

or $\quad -13x_1 = -13x_2$

or $\quad x_1 = x_2$

$\therefore$ f is injective

ii) Surjective



R – {7/3} (domain) R – {4/3} (codomain)

Consider an arbitrary element y in R – {4/3} (codomain)

Let $\quad y = f(x)$

$$y = \frac{4x-5}{3x-7}$$

or $\quad 3xy - 7y = 4x - 5$

or $\quad 3xy - 4x = 7y - 5$

or $\quad x = \dfrac{7y-5}{3y-4}$

$\forall\, y \in R - \{4/3\}$ (codomain) $\exists$ pre image $x \in R - \{7/3\}$ (domain)

$\Rightarrow$ Range of f = codomain

$\Rightarrow$ f is surjective

$\because$ f is injective & surjective both

$\therefore$ f is bijective

$\therefore$ f^{-1} exists.

Let $\quad y = f(x) \quad \Rightarrow x = f^{-1}(y)$

$$y = \frac{4x-5}{3x-7}$$

$\therefore \quad 3xy - 7y = 4x - 5$

$$\Rightarrow x = \frac{7y-5}{3y-4} = f^{-1}(y)$$

$\therefore$ The rule for f^{-1} is

$$f^{-1}(x) = \frac{7x-5}{3x-4}$$

2. $f : R - \left\{\frac{2}{5}\right\} \to R - \left\{\frac{4}{5}\right\}$ defined by $f(x) = \dfrac{4x+3}{5x-2}$ show that the function is bijective and find rule for f^{-1}. **[D-99, M-01]**

Soln. :

$\because$ $f : R - \left\{\frac{2}{5}\right\} \to R - \left\{\frac{4}{5}\right\}$ is defined by $f(x) = \dfrac{4x+3}{5x-2}$

i) Injective :

Consider $\quad f(x_1) = f(x_2)$

$$\frac{4x_1+3}{5x_1-2} = \frac{4x_2+3}{5x_2-2}$$

or $\quad (4x_1+3)(5x_2-2) = (4x_2+3)(5x_1-2)$

or $\quad 20x_1x_2 - 8x_1 + 15x_2 - 6 = 20x_1x_2 - 8x_2 + 15x_1 - 6$

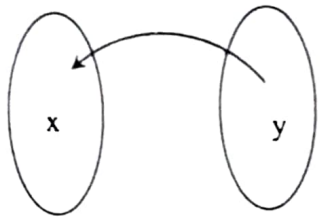
or $\quad -8x_1 - 15x_1 = -8x_2 - 15x_2$

or $\quad -23x_1 = -23x_2$

or $\quad x_1 = x_2$

$\therefore$ f is injective.

ii) surjective :



R – {2/5} (domain) R – {4/5} (codomain)

Consider an arbitrary element y in R – {4/5} (codomain)

Let $y = f(x)$

$$y = \frac{4x+3}{5x-2}$$

or $5xy - 4x = 4x + 3$

or $5xy - 4x = 2y + 3$

or $x(5y-4) = 2y + 3$

or $x = \dfrac{2y+3}{5y-4}$

$\Rightarrow \forall\, y \in R - \left\{\dfrac{4}{5}\right\}$ (codomain)

$\exists$ pre image $x \in R - \left\{\dfrac{2}{5}\right\}$ (domain)

$\Rightarrow$ Range of f = codomain

$\Rightarrow$ f is surjective

$\because$ f is injective and surjective both

$\therefore$ f is bijective

$\therefore$ f^{-1} exist

$y = f(x) \Rightarrow x = f^{-1}(y)$

$\Rightarrow x = \dfrac{2y+3}{5y-4} = f^{-1}(y)$

$\therefore$ The rule for f^{-1} is

$$f^{-1}(x) = \frac{2x+3}{5x-4}$$

3. A function $f : R - \{2\} \to R$ is defined by $f(x) = \dfrac{1}{x-2}$

Is it i) injective ii) surjective iii) bijective

Soln. :

$\because$ $f : R - \{2\} \to R$ is defined by $f(x) = \dfrac{1}{x-2}$

i) Injective :

Consider $f(x_1) = f(x_2)$

$$\frac{1}{x_1 - 2} = \frac{1}{x_2 - 2}$$

or $x_2 - 2 = x_1 - 2$

$\therefore$ $x_1 = x_2$

ii) Surjective :

R – {2} (domain) R (codomain)

Consider an arbitrary element y in R (codomain)

Let $y = f(x)$

$$y = \frac{1}{x-2}$$

or $\quad xy - 2y = 1$

or $\quad xy = 1 + 2y$

or $\quad x = \frac{1}{y} + 2$

For $y = 0 \in R$ (codomain), x is not defined in $R - \{2\}$ (domain)

$\therefore$ Range of $f \neq$ codomain

$\therefore$ f is not surjective

$\therefore$ f is not bijective.

4. Let $f : R \to R$ be a function defined by $f(x) = 2x - 3$.
Prove that it is bijective hence find inverse.

Soln. :

Here $f : R \to R$ is defined by $f(x) = 2x - 3$

i) Injective :

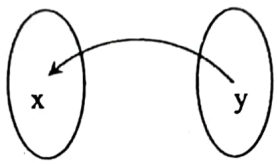
Consider $f(x_1) = f(x_2)$

$2x_1 - 3 = 2x_2 - 3$

$\Rightarrow x_1 = x_2$

$\therefore$ f is injective.

ii) Surjective :



R (domain) R (codomain)

Consider an arbitrary element y in R (codomain)

Let $\quad y = f(x)$

$y = 2x - 3$

or $\quad y + 3 = 2x$

or $\quad x = \frac{y+3}{2}$

$\Rightarrow \forall\, y \in R$ (codomain) $\exists$ pre image $x \in R$ (domain)

$\Rightarrow$ Range of f = codomain

$\Rightarrow$ f is surjective

$\because$ f is injective and surjective both

$\therefore$ f is bijective.

$\therefore$ f^{-1} exists.

$y = f(x) \quad \Rightarrow x = f^{-1}(y)$

$y = 2x - 3$

$x = \frac{y+3}{2} = f^{-1}(y)$

$\therefore$ The rule for f^{-1} is

$f^{-1}(x) = \frac{x+3}{2}$

5. Let $f : R - \{3\} \to R - \{0\}$ is defined by

$f(x) = \frac{1}{x-3}$ then prove that it is bijective and find f^{-1} rule.

Soln. :

Let $f : R - \{3\} \to R - \{0\}$ is defined by $f(x) = \frac{1}{x-3}$

i) Injective :

Consider $f(x_1) = f(x_2)$

$\frac{1}{x_1 - 3} = \frac{1}{x_2 - 3}$

or $\quad x_2 - 3 = x_1 - 3$

$\Rightarrow \quad x_2 = x_1$

$\therefore$ f is injective

ii) surjective :

R – {3} (domain) R – {3} (codomain)

Consider an arbitrary element y in R – {3} (codomain)

$$y = f(x)$$

$$y = \frac{1}{x-3}$$

or $xy - 3y = 1$

or $xy = 1 + 3y$

or $x = \frac{1}{y} + 3$

$\Rightarrow \quad y \in R - \{0\}$ (codomain)

$\exists$ pre image $x \in R - \{3\}$ (domain)

$\Rightarrow$ Range of f = codomain

$\Rightarrow$ f is surjective

$\because$ f is injective and surjective both

$\therefore$ f is bijective

$\therefore$ f^{-1} exist

$y = f(x) \quad \Rightarrow x = f^{-1}(y)$

$$y = \frac{1}{x-3}$$

or $xy - 3y = 1$

or $x = \frac{1}{y} + 3 = f^{-1}(y)$

$\therefore$ The rule for f^{-1} is

$$f^{-1}(x) = \frac{1}{x} + 3$$

6. A function $f : R \to R$ is defined by $f(x) = x^2$

Is it i) injective ii) surjective iii) bijective

Soln. :

Let a function $f : R \to R$ is defined by $f(x) = x^2$

i) Injective :

Consider $f(x_1) = f(x_2)$

$\therefore \quad (x_1)^2 = (x_2)^2$

$\therefore \quad x_1 = \pm x_2$

$\Rightarrow \quad x_1 = x_2$ and $x_1 = -x_2$

$\therefore$ f is not injective

Consider $2, -2 \in R$

Let $x_1 = 2, \; x_2 = -2$

We have $x_1 \neq x_2$

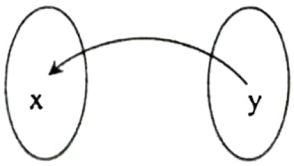
$f(x_1) = f(2) = 4$

$f(x_2) = f(-2) = 4$

$x_1 \neq x_2 \Rightarrow f(x_1) = f(x_2)$

$\therefore$ It is not injective

ii) Surjective :



R (domain) R (codomain)

Consider an arbitrary element y in R (codomain)

Let $y = f(x)$

$y = x^2$

$x = \pm\sqrt{y}$

if y is –ve, x cannot be found in R (domain)

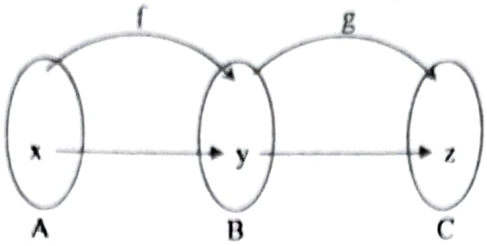
$\therefore$ f is not surjective

It is neither injective nor surjective

$\therefore$ f is not bijective.

Composite Function :

Let $f : A \to B$ and $g : B \to C$ be two functions, then gof is said to be composite function of f and g. if $gof : A \to C$ is a function and is defined as $gof(x) = g\{f(x)\}$.



$y = f(x)$ $\quad$ $z = g(y)$

$z = g\{f(y)\}$

* $fog \neq gof$ (Not commu but associative

* If f is bijec, g is bij gof is also bijective

* If f is bijective, f^{-1} is bij

Note :

$$(fog)^{-1} = g^{-1} o f^{-1}$$

Examples :

1. $f : R \to R$ is defined by $f(x) = x^3$
   $g : R \to R$ is defined by $g(x) = 4x^2 + 1$
   $h : R \to R$ is defined by $h(x) = 7x - 2$
   Find the rule defining
   i) fog $\quad$ ii) gof
   iii) (goh)of $\quad$ iv) go(hof)

[M-01, 04, N-04, M-06]

Q Consider $f(x) = 2x - 3$, $g(x) = x^2 +$ Find a formula for the composition function i) gof and fog

Solu^n i) $gof(x) = g(f(x))$
$= g(2x-3)$
$= (2x-3)^2 + 3(2x-3) + 5$
$= 4x^2 - 6x + 5$

$fog(x) = f(g(x)) = f(x^2+3x+5)$
$= 2(x^2+3x+5) - 3$
$= 2x^2 + 6x + 7$

Soln. :

i) $(fog)(x) = f\{g(x)\}$
$= f(4x^2 + 1)$
$= (4x^2 + 1)^3$

ii) $(gof)(x) = g\{f(x)\}$
$= g(x^3)$
$= 4(x^3)^2 + 1$

Note : $fog \neq gof$

iii) $\{(goh)of\}(x) = [(goh)\{f(x)\}]$
$= (goh)(x^3)$
$= g\{h(x^3)\}$
$= g(7x^3 - 2)$
$= 4(7x^3 - 2)^2 + 1$

iv) $\{go(hof)\}(x) = g\{(hof)(x)\}$
$= g[h\{f(x)\}]$
$= g[h(x^3)]$
$= g(7x^3 - 2)$
$= 4(7x^3 - 2)^2 + 1$

Note : (goh)of = go(hof)

IR – set of Real No's.

2. Let A = B = C = IR and consider, the functions $f : A \to B$ and $g : B \to C$ defined by $f(a) = 2a + 1$ and $g(b) = b/3$ then verify that
   $(gof)^{-1} = f^{-1} o g^{-1}$

Soln. :

Here A = B = C = IR

$f : A \to B$

$\Rightarrow f : R \to R$ is defined as

$f(a) = 2a + 1$

It is bijective

$\therefore$ f^{-1} exists

Let $f(a) = b \Rightarrow a = f^{-1}(b)$

$\therefore$ $2a + 1 = b$

$\therefore$ $a = \dfrac{b-1}{2} = f^{-1}(b)$

$\therefore$ Rule for $f^{-1}(a) = \dfrac{a-1}{2}$

Similarly, here $g : B \to C$

$\Rightarrow$ $g : R \to R$ is defined as

$g(b) = b/3$. It is bijective

let $g(b) = c$

$\Rightarrow$ $b = g^{-1}(c)$

$\therefore$ $\dfrac{b}{3} = c$

$\therefore$ $b = 3c = g^{-1}(c)$

$\therefore$ Rule for $g^{-1}(a) = 3a$

Here $f(a) = 2a + 1$ and $g(a) = b/3$

$$\begin{aligned} \therefore (gof)(a) &= g[f(a)] \\ &= g(2a+1) \\ &= \frac{(2a+1)}{3} \end{aligned}$$

$\because$ f and g are bijective

$\therefore$ gof is bijective

$\therefore$ it is Invertible.

Let, $(gof)(a) = b \Rightarrow a = (gof)^{-1}(b)$

$\therefore$ $\dfrac{2a+1}{3} = b$

$\therefore$ $2a + 1 = 3b$

$\therefore$ $a = \dfrac{3b-1}{2} = (gof)^{-1}(b)$

$\therefore$ Rule for $(gof)^{-1}$ is

$$(gof)^{-1}(a) = \frac{3a-1}{2} \quad \ldots (1)$$

Now, $(f^{-1}og^{-1})(a)$

$$\begin{aligned} &= f^{-1}[g^{-1}(a)] \\ &= f^{-1}(3a) \\ &= \frac{3a-1}{2} \quad \ldots (2) \end{aligned}$$

from (1) and (2) we get,

$\therefore$ $(gof)^{-1} = (f^{-1}og^{-1})$

3. Let A = B = R (set of real no.)
let $f : A \to B$ be given by the formula $f(x) = 2x^3 - 1$ and $g : B \to A$ be given by $g(y) = \sqrt[3]{\frac{1}{2}y + \frac{1}{2}}$, show that f is bijective (i.e. bijective) between A & B & g is bijective between B & A. **[M-03]**

Soln. :

Here, A = B = R

$F : A \to B$

$\Rightarrow$ $f : R \to R$ is defined as

$f(x) = 2x^3 - 1$

Q. $f(x) = 2x + 5$ and $g(x) = 3x - 1$ on R (8)

Find f^{-1}, g^{-1} and verify $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$

Solu^n Let $y = f(x) \Rightarrow x = f^{-1}(y)$

$y = 2x + 5$

$\Rightarrow x = \dfrac{y-5}{2}$

$f^{-1}(x) = \dfrac{x-5}{2}$

Let $y = g(x) \Rightarrow x = g^{-1}(y)$

$y = 3x - 1$

$\Rightarrow x = \dfrac{y+1}{3}$

$\Rightarrow g^{-1}(x) = \dfrac{x+1}{3}$

Now,

$gof(x) = g\{f(x)\} = g\{2x+5\}$

$= 3(2x+5) - 1$

$= 6x + 14$

$f^{-1}og^{-1}(x) = f^{-1}\{g^{-1}(x)\}$

$= f^{-1}\left\{\dfrac{x+1}{3}\right\}$

$= \dfrac{\frac{x+1}{3} - 5}{2} = \dfrac{\frac{x+1-15}{3}}{2}$

$= \dfrac{x-14}{6}$ — (1)

Now

$gof(x) = g\{f(x)\} = g\{2x+5\}$

$= 3(2x+5) - 1$

$= 6x + 14$

Let $y = gof(x) \Rightarrow gof(x) = f^{-1}(y)$

$y = 6x + 14$

$\Rightarrow x = \dfrac{y-14}{6}$

$\Rightarrow (gof)^{-1}(x) = \dfrac{x-14}{6}$ — (2)

(1) = (2) verified

2. $f(x) = -3x + 4$ $\quad g(x) = 5x + 2$

$f^{-1}(x) = \dfrac{4-x}{3}$; $g^{-1}(x) = \dfrac{x-2}{5}$

$gof(x) = -15x + 22$

$(gof)^{-1}(x) = \dfrac{22-x}{15}$

$f^{-1}og^{-1}(x) = f^{-1}\left(\dfrac{x-2}{5}\right)$

$= \dfrac{4-\left(\frac{x-2}{5}\right)}{3} = \dfrac{22-x}{15}$

Total Number students who have attempted CO	122	116	113
no of students scoring more than 50% marks	114	109	111
% of students scoring more than 50% marks	93	94	97

Attainment Level	1	50-59.99%
	2	60-69.99%
	3	70% and above
	-	less than 50 %

**Note: Calculate respective COs using weighted average of all the questions where a particular CO is mapped.

Attained CO1	94.0	3
Attained CO2	93.4	3
Attained CO3		
Attained CO4		
Attained CO5		
Attained CO6	96.5	3

8. $f(x) = 2 - 3x$ $\quad g(x) = x - 9$

$(1) = (2)$

$= \frac{29 - x}{2}$ — (2)

$f' \circ g'(x) = f'\left(\frac{1-x}{4}\right) = 2\left(\frac{1-x}{4} + 2\right)$

$(g \circ f(x))' = \frac{29 - x}{2}$ — (1)

$x = \frac{29 - y}{2}$

$2x = 29 - y$

$y = -2x + 29$

$g \circ f(x) = g\left(\frac{x}{2} - 2\right) = -4\left(\frac{x}{2} - 2\right) + 1 = -2x + 29$

$f'(x) = 2(x + 2)$ $\quad g'(x) = \frac{1-x}{4}$

$x = 2(y + 2)$ $\quad x = \frac{1-y}{4}$

$\frac{x}{2} = y + 2 \Rightarrow y = \frac{x}{2} - 2$ $\quad 4x = 1 - y$

$y = -4x + 1$

$f(x) = y = x \Rightarrow f'(y) = x$ $\quad g(x) = y \Rightarrow g'(y) = x$

$f(x) = \frac{x}{2} - 2$ $\quad g(x) = -4x + 1$

i) Injective :

Consider $f(x_1) = f(x_2)$

$\therefore\ 2x_1^3 - 1 = 2x_2^3 - 1$

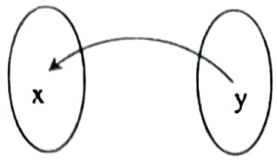
$\therefore\ 2x_1^3 = 2x_2^3$

$\therefore\ x_1^3 = x_2^3$

$\therefore\ x_1 = x_2$

$\therefore$ f is injective

ii) Surjective :



R (domain) R (codomain)

Let $y = f(x)$

$= 2x^3 - 1$

$\therefore\ 2x^3 = y + 1$

$\therefore\ x^3 = \dfrac{y+1}{2}$

$\therefore\ x^3 = \dfrac{y}{2} + \dfrac{1}{2}$

$x = \sqrt[3]{\dfrac{y}{2} + \dfrac{1}{2}}$ (1)

$\therefore\ \forall\, y \in R$ (codomain)

$\exists$ pre image $x \in R$ (domain)

$\Rightarrow$ 'f' is surjective

iii) Bijective :

$\because$ f is injective and surjective both

$\because$ 'f' is bijective

$\Rightarrow f^{-1}$ exists.

Let $y = f(x) \Rightarrow x = f^{-1}(y)$

$= 2x^3 - 1$

$\therefore\ x = \sqrt[3]{\dfrac{y}{2} + \dfrac{1}{2}} = f^{-1}(y)$ (from (1))

$\because$ f is bijective

$\therefore\ f^{-1}$ is bijective

$\because\ f^{-1} = g$ $[\because f^{-1}(y) = g(y)]$

$\therefore$ g is also bijective

4. Define **[M-97]**

$f : N \to N$ by

$$f(n) = \begin{cases} n+1 & \text{if } n \text{ is odd} \\ n-1 & \text{if } n \text{ is even} \end{cases}$$

$g : N \to N$ by

$$g(n) = \begin{cases} n+1 & \text{if } n \text{ is even} \\ 1 & \text{if } n = 1 \\ n-1 & \text{if } n \text{ is odd} \end{cases}$$

Prove that f and g are bijection

Soln :

Part I :

Consider the function f.

Let $f(n_1) = f(n_2)$ n_1 & n_2 both odd.

$\Rightarrow n_1 + 1 = n_2 + 1$

$\Rightarrow n_1 = n_2$

$f(n_1) = f(n_2)$ n_1, n_2 both even

$\Rightarrow n_1 = n_2$

f is one–one i.e. injective

Let $n \in N$ be an arbitrary that may either even or odd

Two cases arises

Case (i) : If n is even then $n - 1$ is odd

$f(n - 1) = n - 1 + 1$

$= n$

$\Rightarrow$ for even n its preimage $(n - 1)$ exists in N.

Case (ii) : If n is odd then $n + 1$ is even

$f(n + 1) = n + 1 - 1$

$= n$

$\Rightarrow \forall$ n is an image of $n + 1$

$\Rightarrow$ Range of f = codomain

$\therefore$ f is surjective

Hence f is "Bijection"

Part II :

To prove that g is bijection

$g(1) = 1$ (by def.)

Let $g(n_1) = g(n_2)$, n_1, n_2 both even

$\Rightarrow n_1 + 1 = n_2 + 1$

$\Rightarrow n_1 = n_2$

Let $g(n_1) = g(n_2)$ n_1, n_2 both odd but n_1 and n_2 is not equal to 1.

$\Rightarrow n_1 - 1 = n_2 - 1$

$\Rightarrow n_1 = n_2$

$\therefore$ f is injective.

Consider an arbitrary $n \in N$ that may either even or odd but not equal to 1.

Case (i) : If n is even then $n + 1$ is odd. $n > 1$

$\Rightarrow g(n + 1) = (n + 1) - 1$

$= n$

$\Rightarrow$ n is image of $n + 1 \in N$

Case (ii) : If n is odd then $n - 1$ is even $n > 1$

$\Rightarrow g(n - 1) = (n - 1) + 1$

$= n$

$\Rightarrow$ n is image of $n - 1 \in N$

$\therefore$ $n \in N$ has preimage in N

$\Rightarrow$ g is surjective.

$\because$ g is injective and surjective both.

$\therefore$ g is bijective.

[M-97] **5.** Determine whether the function is one–one, onto [D-98, N-04]

$f : I \to I$

$$f(j) = \begin{cases} j/2 & \text{if } j \text{ is even} \\ (j-1)/2 & \text{if } j \text{ is odd} \end{cases}$$

where I is the set of all integers.

Soln. :

If we take $j = 2$ $f(2) = 2/2 = 1$

$j = 3$ $f(3) = 3-1/2 = 1$

$2 \neq 3 \Rightarrow f(2) = f(3) \Rightarrow$ f is not one–one

Consider an arbitrary $j \in I$

$\Rightarrow$ j is either even or odd

Case (i) : j is even

$\Rightarrow 2j \in Z$

$\because f(2j) = 2j/2$

$= j$

$\therefore$ j has preimage 2j in I.

Case (ii) : j is odd

$\Rightarrow$ $2j + 1$ is also odd

$\because$ $f(2j+1) = \dfrac{(2j+1)-1}{2}$

$= j$

$\therefore$ j has preimage in I

$\therefore$ Each element of I has preimage in I

$\Rightarrow$ Range = codomain

$\Rightarrow$ f is onto.

Pigeonhole Principle

If 'n' pigeons are assigned to 'm' pigeon holes ($m < n$) then at least one pigeon hole must have two pigeons.

1. If 13 students are selected from a class, prove that atleast two of them must have their birthday on the same month of a year.

Soln. :

Here, 13 students (13 pigeons) are selected from a class.

There are 12 months (12 pigeonholes) in a year.

Since no. of pigeonholes is less than the no. of pigeons,

$\therefore$ Pigeonhole principle is applicable.

By Pigeonhole principle, atleast one pigeonhole must contain two pigeons.

$\Rightarrow$ At least two students must have their birthday on the same month of the year.

2. Prove that if 8 people are assembled in a room then 2 of them are born on the same day of a week.

Soln. :

There are 7 days (7 pigeonholes) in a week and 8 people (8 pigeon) are assembled in a room.

Since no. of pigeonholes is less than the no. of pigeons,

$\therefore$ Pigeonhole principle is applicable.

By pigeonhole principle, one pigeonhole must contain atleast two pigeons.

$\Rightarrow$ Atleast two people must have borned on the same day of a week.

3. If 11 numbers are chosen from a set = {1, 2, ... 20} prove that one of them is multiple of other. **[M-05]**

Soln. :

Any natural number can be expressed $b = 2^k . m$

where 'm' is known as odd part of 'n'. and $k \geq 0$.

By this way we find there are 10 odd parts in the set {1, 2, ... 20}

$\because$ there are 10 odd parts (10 pigeon holes) and chosen numbers are 11 (11 pigeon).

$\Rightarrow$ Pigeonhole principle is applicable

By pigeon hole principle one pigeonhole must have atleast two pigeons

$\therefore$ Two natural numbers must have same odd part

Let n_1 & n_2 have same odd part 'n'

$n_1 = 2^{k_1} m$

$n_2 = 2^{k_2} m$

there are two possibilities

i) $k_1 < k_2$

$\Rightarrow n_1 \mid n_2$

ii) $k_1 > k_2$

$\Rightarrow n_2 \mid n_1$

either way one is multiple of other.

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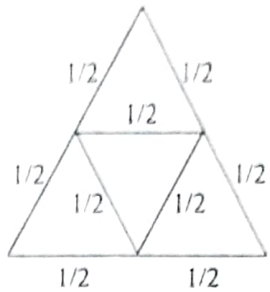
4. Let T is an equilateral triangle of side 1 unit. Prove that if 5 points are chosen inside or on the triangle, then two of them are not more than 1/2 unit apart.

Soln. :

From this figure, we find there are four small equilateral triangles of sides 1/2 unit. Since we are choosing 5 points (5 pigeons) and there are 4 small triangles (4 pigeon holes)



Since no. of pigeonholes is less than the no. of pigeons.
Pigeonhole principle is applicable
By pigeonhole principle, one pigeonhole must have atleast two pigeons
∴ One small triangle must have two points
∵ Sides of each small triangle is 1/2 unit,
∴ Two of them can not be more than 1/2 unit apart.

5. If 5 numbers are chosen from a set {1, 2, 3, ..., 8} then prove that two of them will add up 9.

Soln. :

The numbers of the set {1, 2, 3 ... 8}, can be grouped in pairs such that the sum of each pair = 9, i.e., (1, 8), (2, 7), (3, 6), (4, 5).
∵ There are four groups (4 pigeonholes) and 5 numbers (5 pigeons) are chosen.
⇒ Pigeonhole principle is applicable.
By pigeonhole principle one pigeonhole must have atleast two pigeons.
⇒ Atleast one group contains 2 numbers.
∴ Atleast two numbers will add up 9.

6. Shirts numbered consecutively from 1 to 20 are worn by the 20 members of a bowling league. When any 3 of these members are chosen to be a team, the sum of their shirt numbers is used as a code number for the team. Show that if any 8 of the 20 are selected, then from these 8 we may form at least two different teams having the same code number. **[M-02]**

Soln. :

Since the code number of a team is sum of their shirt numbers and shirt numbers are taken form a set {1, 2, ...20}, therefore one of the lowest code number is 1 + 2 + 3 = 6 and highest code numbers is 18 + 19 + 20 = 57. By this way, we find total number of code numbers is equal to 52. (including code numbers 6 and 57).
∵ 8 are selected from 20 and a team contains three members, therefore number of teams of three members out of 8 is equal to $^8C_3 = 56$.
Since, total code number is less than the number of prepared teams (52 < 56).
∴ Pigeon hole principle is applicable.
By Pigeonhole principle one pigeonhole must have atleast two pigeons.
∴ Atleast two different teams have the same code number.

Extended Pigeonhole Principle :

If 'n' pigeons are assigned to 'm' pigeonholes (km < n), then at least one pigeonhole must have $\left\lfloor \frac{n-1}{m} \right\rfloor + 1$ pigeons.

Note : $\left\lfloor \frac{m}{n} \right\rfloor$ = the largest integer which is less or equal to $\left(\frac{m}{n}\right)$.

e.g., $\left\lfloor \frac{29}{5} \right\rfloor = 5$, $\left\lfloor \frac{35}{7} \right\rfloor = 5$, $\left\lfloor \frac{7}{3} \right\rfloor = 2$.

1. Show that if 30 dictionaries in a library contain a total of 61,327 pages, then one of the dictionaries must have at least 2045 pages.

Soln :

There are 30 dictionaries (30 pigeonholes) and these contain 61,327 pages (61,327 pigeons).
∵ 30 << 61,327

$\therefore$ Extended pigeon hole principle is applicable.

By extended pigeon hole principle, one pigeon hole must have atleast $\left\lfloor \frac{n-1}{m} \right\rfloor + 1$ pigeons.

Here, n = 61327, m = 30.

$\therefore \left\lfloor \frac{n-1}{m} \right\rfloor + 1 = 2044 + 1$

$= 2045$ pigeons.

$\therefore$ One dictionary must have atleast 2045 pages.

2. If 30 people are assembled in a room. Show that, 5 of them must have their birthday on the same day of a week.

Soln :

There are 7 days (7 pigeonholes) in a week and 30 people (30 pigeons) are assembled in a room.

$\because$ Number of pigeonholes << number of pigeons (7 << 30).

$\therefore$ Extended pigeonhole principle is applicable.

Here n = 30, m = 7.

$\therefore \left\lfloor \frac{n-1}{m} \right\rfloor + 1 = 4 + 1 = 5$ pigeons

$\therefore$ 5 of them must have their birthday on the same day of a week.

3. Prove that if 7 colours are used to paint 50 bicycles, then 8 of them must have same colour : **[M-01, N-04]**

Soln :

There are 50 bicycles (50 pigeons) and 7 colours (7 pigeonholes).

$\because$ Number of pigeonholes << number of pigeons (7 << 50).

$\therefore$ Extended pigeonhole principle is applicable

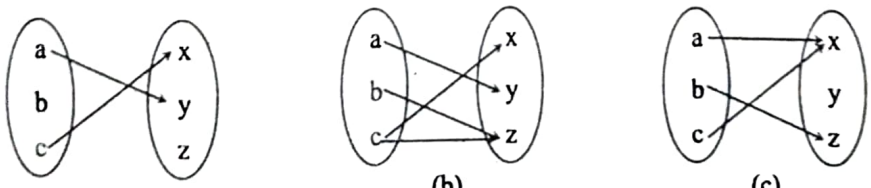
n = 50, m = 7.

$\left\lfloor \frac{50-1}{7} \right\rfloor + 1 = 7 + 1 = 8$ pigeons.

$\therefore$ 8 of them must have same colours.

Graded Questions

1. Define the following terms with example :
   (a) Injective function or one to one function
   (b) Surjective function or onto function
   (c) Bijection / bijective function
   (d) Inverse function
2. State whether or not each diagram in fig. below defines a function from A = {a, b, c} into B = {x, y, z}



(a) (b) (c)

3. Let X = {1, 2, 3, 4}. Determine whether or not each relation below is a function from X into X.
   (a) f = {(2, 3), (1, 4), (2, 1), (3, 2), (4, 4)}
   (b) g = {(3, 1), (4, 2), (1, 1)}
   (c) h = {(2, 1), (3, 4), (1, 4), (2, 1), (4, 4)}

4. If mapping $f : A \to B$ is one to one and onto, prove that inverse mapping $f^{-1} : B \to A$ is also one to one and onto. **[M-04, 05]**

5. Let $N = \{0, 1, 2, \ldots\ldots n-1\}$ where n is even. **[M-98]**
   Define $f : N \to N$ by –
   $f(i) = 2i \qquad 0 \le i \le n/2-1$
   $\quad = (2i+1) \bmod n \qquad n/2 \le i \le n-1$
   Is f a bijection ? Explain.
   If it is, find $f^{-1}(i)$.

6. Let $f : R \to (7/5) \to R - \left(\frac{2}{5}\right)$ be defined by $f(x) = \frac{2x-3}{5x-7}$. **[M-02]**
   Prove that it is a bijection. Hence find f^{-1}.

7. If f and g be the functions from set of integers to set of integers defined by $f(x) = 2x + 3$, $g(x) = 3x + 2$. Find fog and gof. **[D-02]**

8. Let the functions f and g be defined by $f(x) = 2x + 1$ and $g(x) = x^2 - 2$. Find the formula defining the composition function gof.

9. Show that function $f(x) = ax + b$ from R to R is invertible. where a and b are constants with $a \neq 0$. Find the inverse of f. **[M-03]**

10. Explain the bijective function with suitable example. **[D-01]**
    Let $f : R \to R$ defined as $f(x) = 2x^3 - 7$
    $g : R \to R$ defined as $g(x) = 3x^2$
    $h : R \to R$ defined as $h(x) = 5x + 4$
    Find the rule of defining (i) (gof) oh (ii) fo(hog).

11. Let A = B = C = R and let $f : A \to B$ and $g : B \to C$ be defined by $f(a) = a - 1$ and $g(b) = b^2$ find (i) $(gof)_{(x)}$ (ii) $(fof)_{(y)}$ (iii) $(gog)_{(y)}$ **[D-03]**

12. Let A = B = C = R and let $f : A \to B$, $g : B \to C$ be defined by $f(a) = a + 1$ and $g(b) = b^2 + 2$. Find – **[D-01, N-04]**
    (i) (fog) (2) (ii) (go f) (x) (iii) (fog) (x) (iv) (gog) (y) (v) fo(gog) (x)

13. Let $f : A \to B$ be one to one and onto then prove that – **[D-02, 03]**
    $f^{-1} 0 = f = I_A$
    $f 0 f^{-1} = I_B$ where I_A and I_B are identical mapping on set A and set B.

Pigeonhole Principle

Type I

1. Prove that if any 14 nos. from 1 to 25 are chosen, then one of them is a multiple of another. **[D-01]**
2. Prove that if 6 numbers are chosen from {1, 2, ... 11} then two of them will add up twelve.
3. If any seven points are chosen from a regular hexagon whose sides are of length 1. Prove that two of them must be no further apart than 1 unit.
4. Show that if any eight positive integers are chosen, two of them will have the same remainder when divided by 7.

Type II

1. If 62 students are selected from a class then atleast 6 of them must have their Birthday on the same month of a year.
2. How many friends must you have to guarantee that at least five of them will have birthdays in the same month? **[M-04]**
3. What is the minimum number of student required in a discrete structure class to be sure that at least six will receive the same grade, if there are five possible grades A, B, C, D and F ? **[D-03]**

□□□□□□